HITESH NANDAKUMAR

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EDUCATION

California Polytechnic State University, San Luis Obispo
Bachelor of Science in Computer Engineering , *Concentration in Embedded Systems*
Minor in Ethics, Public Policy, Science, and Technology
Expected June 2028 • GPA: 3.5

TECHNICAL SKILLS

Hardware & Digital Design: SystemVerilog • RTL Design • Finite State Machines • Datapath & Control Architecture • RISC-V (RV32I) • Timing Analysis • Simulation & Waveform Debugging • Vivado • Basys 3 (Artix-7)
Embedded & Electronics: Sensor Interfacing • Circuit Debugging • Oscilloscope • Logic Analyzer • Digital Multimeter • Signal Generator • LTspice
Programming: C/C++ • Python • Java • C# • JavaScript • HTML/CSS
Development Tools: Git • GitHub • VS Code • IntelliJ • Eclipse • UML & System Architecture Design

ENGINEERING PROJECTS

32-bit RISC-V Microcontroller (OTTER MCU)
SystemVerilog • Computer Architecture • Digital Systems Design
• Implemented a 32-bit RV32I processor in SystemVerilog from ISA specification, designing datapath and control logic
• Developed test cases and performed RTL simulation with waveform debugging to validate functional correctness
• Executed timing analysis and structured regression testing across instruction formats and memory-mapped I/O

FSK IR Communication System
Analog Circuit Design • IR Communication • Signal Processing
• Designed and built an FSK-based IR communication link with defined signal integrity and frequency discrimination requirements
• Simulated multi-stage analog front-end (TIA, comparator, filtering) in LTspice to validate gain, and noise margins
• Conducted end-to-end validation testing to ensure consistent ASCII data transmission

IEEE Technical Report – Low-Voltage Sensing and Indication Circuit
Analog Circuit Design • Experimental Validation • Technical Documentation
• Designed and validated an undervoltage detection circuit using datasheet-driven component and tolerance analysis
• Developed repeatable voltage sweep testing (7–10 V) to characterize trip accuracy and switching behavior
• Quantified performance (8.02 V trip, −6.3 mV error) and documented traceable validation results in IEEE format

LEADERSHIP & EXPERIENCE

Resident Advisor (RA)
California Polytechnic State University, San Luis Obispo **| August 2025 – Present**
• Managed conflict resolution and community initiatives in a **high-responsibility residential environment**, supporting 50+ undergraduate residents

Founder & Operator — *Slick and Quick Car Care* Automotive Detailing
Independent Business, Livermore CA **| August 2022 – Present**
• Operated a **customer-facing service business**, managing scheduling and customer service along with a full-time engineering curriculum